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IMPROVEMENTS OF TWO INEQUALITIES IN AN ACUTE TRIANGLE

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Abstract. In this paper, we obtain improvements of two known inequalities in an acute triangle. We also propose three conjectured inequalities as open problems.

1. Introduction and results

In 1996, the author proposed the following conjecture (as the problem shc 15(h) in [11]): If $\triangle ABC$ is an acute triangle then

$$\sum (m_b + m_c)^2 \geq 4s^2, \tag{1.1}$$

where m_a, m_b, m_c and s are the medians and the semiperimeter of $\triangle ABC$ respectively, the symbol $\sum$ denotes the cyclic sum.

X. G. Chu [14] first proved inequality (1.1). The author of this paper provided two new proofs in the recent paper [7].

By the well-known identity:

$$\sum m_a^2 = \frac{3}{4}\sum a^2, \tag{1.2}$$

where a, b, c are the sides of triangle ABC. We can give several equivalent forms of inequality (1.1) (cf. [7]). One of them is

$$\sum m_b m_c \geq \sum bc - \frac{1}{4}\sum a^2. \tag{1.3}$$

For any triangle ABC, X. G. Chu and X. Z. Yang [15] gave the following inequality without proof:

$$\sum m_a \leq \frac{2s^2}{\sqrt{\sum bc}}, \tag{1.4}$$

which was proved and used by the author in the recent paper [6].

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Recently, inspired by inequality (1.4), the author found that for the acute triangle the following inequality holds:

$$\sum m_b m_c \geq \frac{4s^4}{3\sum a^2}, \tag{1.5}$$

which actually improves inequality (1.3). By identity (1.2), we see that this inequality can be equivalently stated as the following:

Theorem 1.1. *In the acute triangle ABC the following inequality holds:*

$$\sum m_b m_c \sum m_a^2 \geq s^4, \tag{1.6}$$

with equality if and only if triangle ABC is equilateral.

It is easy to verify the following identity:

$$\frac{4s^4}{3\sum a^2} = \sum bc - \frac{1}{4}\sum a^2 + \frac{\left(\sum a^2 - \sum bc\right)^2}{3\sum a^2}, \tag{1.7}$$

which shows that inequality (1.5) is stronger than inequality (1.3).

In [14], X. G. Chu also proved the following acute triangle inequality:

$$\sum m_b m_c \geq \frac{1}{3}\sum r_a \sum h_a, \tag{1.8}$$

where h_a, h_b, h_c are the altitudes of $\triangle ABC$ and r_a, r_b, r_c the radii of excircles of $\triangle ABC$.

We find that using the method to prove inequality (1.6), one can prove the following improvement of inequality (1.8):

Theorem 1.2. *In the acute triangle ABC the following inequality holds:*

$$\sum m_b m_c \geq \frac{1}{12}\left(\sum h_a + \sum r_a\right)^2, \tag{1.9}$$

with equality if and only if triangle ABC is equilateral.

The purpose of this paper is prove Theorem 1.1 and Theorem 1.2. We also propose three conjectures as open problems.

2. Some lemmas

In this section, we give some lemmas which will be used in the proof of Theorem 1.1 and 1.2.

In what follows, we denote by R, r the circumradius and inradius of triangle ABC, respectively.

For any triangle ABC, we have the following well-known identities:

$$\sum a = 2s, \tag{2.1}$$

$$abc = 4Rrs, \tag{2.2}$$

$$\sum bc = s^2 + 4Rr + 3r^2. \tag{2.3}$$

By using these three basis identities, one can prove the following Lemma 2.1-2.3.

Lemma 2.1. *In the triangle ABC the following identities hold:*

$$\sum a^2 = 2s^2 - 8Rr - 2r^2, \tag{2.4}$$

$$\sum a^3 = 2s^3 - (12Rr + 6r^2)s, \tag{2.5}$$

$$\sum a^4 = 2s^4 - 4(4R + 3r)rs^2 + 2(4R + r)^2r^2, \tag{2.6}$$

$$\sum a^5 = 2s^5 - 20(R + r)rs^3 + 10(2R + r)(4R + r)r^2s, \tag{2.7}$$

$$\begin{aligned}\sum a^6 =& 2s^6 - 6(4R + 5r)rs^4 + 6(24R^2 + 24Rr + 5r^2)r^2s^2 \\ & - 2(4R + r)^3r^3,\end{aligned} \tag{2.8}$$

$$\begin{aligned}\sum a^7 =& 2s^7 - 14(2R + 3r)rs^5 + 14(16R^2 + 20Rr + 5r^2)r^2s^3 \\ & - 14(2R + r)(4R + r)^2r^3s,\end{aligned} \tag{2.9}$$

$$\begin{aligned}\sum a^8 =& 2s^8 - 8(4R + 7r)rs^6 + 20(16R^2 + 24Rr + 7r^2)r^2s^4 \\ & - 8(4R + r)(32R^2 + 32Rr + 7r^2)r^3s^2 + 2(4R + r)^4r^4,\end{aligned} \tag{2.10}$$

$$\begin{aligned}\sum a^9 =& 2s^9 - 36(R + 2r)rs^7 + 36(12R^2 + 21Rr + 7r^2)r^2s^5 \\ & - 12(160R^3 + 240R^2r + 105Rr^2 + 14r^3)r^3s^3 \\ & + 18(2R + r)(4R + r)^3r^4s,\end{aligned} \tag{2.11}$$

$$\begin{aligned}\sum a^{10} =& 2s^{10} - 10(4R + 9r)rs^8 + 140(2R + 3r)(2R + r)r^2s^6 \\ & - 20(160R^3 + 280R^2r + 140Rr^2 + 21r^3)r^3s^4 \\ & + 10(40R^2 + 40Rr + 9r^2)(4R + r)^2r^4s^2 \\ & - 2(4R + r)^5r^5,\end{aligned} \tag{2.12}$$

$$\begin{aligned}\sum a^{11} =& 2s^{11} - 22(2R + 5r)rs^9 + 44(16R^2 + 36Rr \\ & + 15r^2)r^2s^7 - 308(2R + r)(8R^2 + 12Rr \\ & + 3r^2)r^3s^5 + 22(4R + r)(160R^3 + 240R^2r \\ & + 108Rr^2 + 15r^3)r^4s^3 - 22(2R + r)(4R + r)^4r^5s,\end{aligned} \tag{2.13}$$

$$\begin{aligned}\sum a^{12} =& 2s^{12} - 12(4R + 11r)rs^{10} + 18(48R^2 + 120Rr \\ & + 55r^2)r^2s^8 - 56(128R^3 + 288R^2r + 180Rr^2 \\ & + 33r^3)r^3s^6 + 6(4480R^4 + 8960R^3r + 6048R^2r^2 \\ & + 1680Rr^3 + 165r^4)r^4s^4 - 12(48R^2 + 48Rr \\ & + 11r^2)(4R + r)^3r^5s^2 + 2(4R + r)^6r^6,\end{aligned} \tag{2.14}$$

$$\begin{aligned}\sum a^{13} =& 2s^{13} - 52(R + 3r)rs^{11} + 130(8R^2 + 22Rr + 11r^2)r^2s^9 \\ & - 312(32R^3 + 80R^2r + 55Rr^2 + 11r^3)r^3s^7 \\ & + 26(1792R^4 + 4032R^3r + 3024R^2r^2 + 924Rr^3 \\ & + 99r^4)r^4s^5 - 52(112R^3 + 168R^2r + 77Rr^2 \\ & + 11r^3)(4R + r)^2r^5s^3 + 26(2R + r)(4R + r)^5r^6s,\end{aligned} \tag{2.15}$$

$$\begin{aligned}\sum a^{14} =&2s^{14} - 14(4R+13r)rs^{12} + 154(8R^2+24Rr+13r^2)r^2s^{10}\\ &- 42(320R^3+880R^2r+660Rr^2+143r^3)r^3s^8 + 42(1792R^4\\ &+ 4480R^3r+3696R^2r^2+1232Rr^3+143r^4)r^4s^6\\ &- 14(4R+r)(3584R^4+7168R^3r+4928R^2r^2+1408Rr^3\\ &+ 143r^4)r^5s^4 + 14(56R^2+56Rr+13r^2)(4R+r)^4r^6s^2\\ &- 2(4R+r)^7r^7. \end{aligned} \tag{2.16}$$

The above identities (2.4)-(2.6) can be found in the monograph [10]. The identities (2.7)-(2.16) have been proved by the author in [3] and [4].

Lemma 2.2. *In the triangle ABC the following identities hold:*

$$\sum b^2c^2 = s^4 - 2r(4R-r)s^2 + r^2(4R+r)^2, \tag{2.17}$$

$$\sum b^3c^3 = s^6 - 3(4R-r)rs^4 + 3r^4s^2 + (4R+r)^3r^3, \tag{2.18}$$

$$\begin{aligned}\sum b^6c^6 =&s^{12} - 6(4R-r)rs^{10} + 3(48R^2-24Rr+5r^2)r^2s^8\\ &- 4(32R^3-24R^2r+12Rr^2-5r^3)r^3s^6\\ &+ 3(16R+5r)r^7s^4 + 6(4R+r)^3r^7s^2\\ &+ (4R+r)^6r^6, \end{aligned} \tag{2.19}$$

$$\begin{aligned}\sum b^9c^9 =&s^{18} - 9(4R-r)rs^{16} + 36(12R^2-6Rr+r^2)r^2s^{14}\\ &- 12(160R^3-120R^2r+42Rr^2-7r^3)r^3s^{12}\\ &+ 18(128R^4-128R^3r+72R^2r^2-28Rr^3+7r^4)r^4s^{10}\\ &+ 126r^{10}s^8 + 12(60R^2+42Rr+7r^2)r^{10}s^6\\ &+ 36(2R+r)(4R+r)^3r^{10}s^4 + 9(4R+r)^6r^{10}s^2\\ &+ (4R+r)^9r^9. \end{aligned} \tag{2.20}$$

Proof. The proofs of identities (2.17)-(2.19) can be found in [4]. In addition, it is easy to obtain

$$\sum b^9c^9 = \sum b^3c^3 \sum b^6c^6 + (abc)^3\left(\sum a^9 - \sum a^3 \sum a^6\right). \tag{2.21}$$

With the help of software Maple, using identities (2.2), (2.5), (2.8), (2.11), (2.18) and (2.19), we immediately obtain identity (2.20).

Lemma 2.3. *In the triangle ABC the following identities hold:*

$$\begin{aligned}\sum a^{15} =&2s^{15} - 30(2R+7r)rs^{13} + 30(48R^2+156Rr+91r^2)r^2s^{11}\\ &- 110(160R^3+480R^2r+390Rr^2+91r^3)r^3s^9\\ &+ 90(1280R^4+3520R^3r+3168R^2r^2+1144Rr^3\\ &+ 143r^4)r^4s^7 - 6(64512R^5+161280R^4r+147840R^3r^2\\ &+ 63360R^2r^3+12870Rr^4+1001r^5)r^5s^5 + 10(896R^3\\ &+ 1344R^2r+624Rr^2+91r^3)(4R+r)^3r^6s^3\\ &- 30(2R+r)(4R+r)^6r^7s, \end{aligned} \tag{2.22}$$

$$\begin{aligned}\sum a^{16} =&2s^{16} - 16(4R + 15r)rs^{14} + 104(16R^2 + 56Rr + 35r^2)r^2s^{12}\\ &- 176(128R^3 + 416R^2r + 364Rr^2 + 91r^3)r^3s^{10}\\ &+ 132(1280R^4 + 3840R^3r + 3744R^2r^2 + 1456Rr^3\\ &+ 195r^4)r^4s^8 - 16(43008R^5 + 118272R^4r + 118272R^3r^2\\ &+ 54912R^2r^3 + 12012Rr^4 + 1001r^5)r^5s^6 + 8(10752R^4\\ &+ 21504R^3r + 14976R^2r^2 + 4368Rr^3 + 455r^4)(4R + r)^2r^6s^4\\ &- 16(8R + 5r)(8R + 3r)(4R + r)^5r^7s^2 + 2(4R + r)^8r^8, \end{aligned} \tag{2.23}$$

$$\begin{aligned}\sum a^{17} =&2s^{17} - 68(R + 4r)rs^{15} + 476(4R^2 + 15Rr + 10r^2)r^2s^{13}\\ &- 884(32R^3 + 112R^2r + 105Rr^2 + 28r^3)r^3s^{11}\\ &+ 748(320R^4 + 1040R^3r + 1092R^2r^2 + 455Rr^3 + 65r^4)r^4s^9\\ &- 748(1536R^5 + 4608R^4r + 4992R^3r^2 + 2496R^2r^3\\ &+ 585Rr^4 + 52r^5)r^5s^7 + 68(4R + r)(10752R^5 + 26880R^4r\\ &+ 24960R^3r^2 + 10920R^2r^3 + 2275Rr^4 + 182r^5)r^6s^5\\ &- 68(192R^3 + 288R^2r + 135Rr^2 + 20r^3)(4R + r)^4r^7s^3\\ &+ 34(2R + r)(4R + r)^7r^8s, \end{aligned} \tag{2.24}$$

$$\begin{aligned}\sum a^{18} =&2s^{18} - 18(4R + 17r)rs^{16} + 360(6R^2 + 24Rr + 17r^2)r^2s^{14}\\ &- 2184(2R + r)(8R^2 + 26Rr + 17r^2)r^3s^{12}\\ &+ 5148(2R + r)(32R^3 + 96R^2r + 78Rr^2 + 17r^3)r^4s^{10}\\ &- 396(4608R^5 + 14976R^4r + 17472R^3r^2 + 9360R^2r^3\\ &+ 2340Rr^4 + 221r^5)r^5s^8 + 24(236544R^6 + 709632R^5r\\ &+ 823680R^4r^2 + 480480R^3r^3 + 150150R^2r^4 + 24024Rr^5\\ &+ 1547r^6)r^6s^6 - 360(2R + r)(192R^3 + 288R^2r + 126Rr^2\\ &+ 17r^3)(4R + r)^3r^7s^4 + 18(72R^2 + 72Rr\\ &+ 17r^2)(4R + r)^6r^8s^2 - 2(4R + r)^9r^9. \end{aligned} \tag{2.25}$$

Proof. It is easy to obtain

$$\sum a^{15} = \sum a^6 \sum a^9 + (abc)^3 \sum b^3c^3 - \sum a^3 \sum b^6c^6. \tag{2.26}$$

Making us of Maple and identities (2.2), (2.5), (2.8), (2.11), (2.18) and (2.19), we easily get identity (2.22).

We easily get

$$\sum a^{16} = \sum a^3 \sum a^{13} + (abc)^3 \sum a^7 - \sum b^3c^3 \sum a^{10}. \tag{2.27}$$

Then using (2.2), (2.5), (2.9), (2.12), (2.18), identity (2.23) follows immediately.

Again, it is easy to get

$$\sum a^{17} = \sum a^6 \sum a^{11} + (abc)^5 \sum bc - \sum a^5 \sum b^6c^6. \tag{2.28}$$

Using (2.2), (2.3), (2.7), (2.8), (2.13), (2.19) and simplifying gives identity (2.24).

Finally, note that

$$\sum a^{18} = \left(\sum a^9\right)^2 - 2\sum b^9c^9.$$

By using (2.11) and (2.20), we get (2.25). This completes the proof of Lemma 2.3.

Lemma 2.4. *In the triangle ABC, let*

$$M_a = \frac{1}{8}(r_b + r_c + 2h_a) + \frac{2m_a^2}{r_b + r_c + 2h_a},$$
$$M_b = \frac{1}{8}(r_c + r_a + 2h_b) + \frac{2m_b^2}{r_c + r_a + 2h_b},$$
$$M_c = \frac{1}{8}(r_a + r_b + 2h_c) + \frac{2m_c^2}{r_a + r_b + 2h_c}.$$

Then

$$\sum \frac{1}{M_a} = \frac{8N_0}{M_0}, \tag{2.29}$$

where

$$\begin{aligned}
N_0 =&(29R + 98r)s^6 + (580R^3 + 2284R^2r + 619Rr^2\\
&+ 638r^3)s^4 + (576R^5 - 4960R^4r - 5804R^3r^2\\
&+ 5056R^2r^3 + 3199Rr^4 + 278r^5)s^2\\
&- (4R - 3r)(28R^2 - 5Rr - 2r^2)(4R + r)^3r > 0,\\
M_0 =&9s^8 + (280R^2 + 248Rr + 1252r^2)s^6 + (1680R^4\\
&+ 5280R^3r + 23184R^2r^2 - 7320Rr^3 + 4534r^4)s^4\\
&+ (1024R^6 - 25600R^5r - 97088R^4r^2 - 68512R^3r^3\\
&+ 10712R^2r^4 + 10984Rr^5 + 1252r^6)s^2\\
&- (16R^2 - 4Rr - r^2)(4R - 3r)^2(4R + r)^3r > 0.
\end{aligned}$$

Proof. We set

$$K_a = \left[(r_c + r_a + 2h_b)^2 + 16m_b^2\right]\left[(r_a + r_b + 2h_c)^2 + 16m_c^2\right],$$
$$K_b = \left[(r_a + r_b + 2h_c)^2 + 16m_c^2\right]\left[(r_b + r_c + 2h_a)^2 + 16m_a^2\right],$$
$$K_c = \left[(r_b + r_c + 2h_a)^2 + 16m_a^2\right]\left[(r_c + r_a + 2h_b)^2 + 16m_b^2\right].$$

Then we have

$$\sum \frac{1}{M_a} = \frac{8\sum(r_b + r_c + 2h_a)K_a}{\prod\left[(r_b + r_c + 2h_a)^2 + 16m_a^2\right]}, \tag{2.30}$$

where $\prod$ denotes the cyclic product.

Next, we first calculate $\sum(r_b + r_c + 2h_a)K_a$. With the help of Maple, using $r_a = S/(s-a), h_a = 2S/a, s = (a+b+c)/2$, the median formula

$$m_a^2 = \frac{1}{2}(b^2 + c^2) - \frac{1}{4}a^2 \tag{2.31}$$

and Heron's formula

$$S=\sqrt{s(s-a)(s-b)(s-c)}, \tag{2.32}$$

one can easily obtain the following identity:

$$\sum(r_b+r_c+2h_a)K_a=\frac{4rsK_0}{\prod a^2(b+c-a)^2}, \tag{2.33}$$

where

$$\begin{aligned}K_0=&4\sum a^{15}+4\sum a\sum a^{14}-45\sum a^2\sum a^{13}+18\sum a^3\sum a^{12}\\&+182\sum a^4\sum a^{11}-137\sum a^5\sum a^{10}-326\sum a^6\sum a^9\\&+302\sum a^7\sum a^8+abc\Big(642\sum a^{12}-354\sum a\sum a^{11}\\&+160\sum a\sum a\sum a^{10}+107\sum a^3\sum a^9-500\sum a^4\sum a^8\\&-77\sum a^5\sum a^7+724\sum b^6c^6\Big)+(abc)^2\Big(913\sum a^9\\&-1044\sum a^2\sum a^7-170\sum a^4\sum a^5-522\sum a\sum a^8\\&+1159\sum a^3\sum a^6\Big)+(abc)^3\Big(1800\sum b^3c^3+1940\sum a^6\\&-696\sum a\sum a^5-552\sum a^2\sum a^4\Big)+(abc)^4\Big(2105\sum a^3\\&-327\sum a\sum a^2\Big)+1554(abc)^5.\end{aligned}$$

With the help of Maple, applying the previous identities (2.1)-(2.3) and identities given in Lemma 2.1-2.3, simplifying gives us the following identity:

$$K_0=256r^5s^3N_0, \tag{2.34}$$

where N_0 is the same as in Lemma 2.4. Also, it is clear that $N_0>0$ from (2.33) and (2.34).

Using $abc=4Rrs$ and the following known identity

$$\prod(b+c-a)=8sr^2, \tag{2.35}$$

we get

$$\prod a^2(b+c-a)^2=1024R^2r^6s^4. \tag{2.36}$$

Thus, it follows from (2.33), (2.34) and (2.36) that

$$\sum(r_b+r_c+2h_a)K_a=\frac{N_0}{R^2}. \tag{2.37}$$

We now calculate $\prod\left[(r_b+r_c+2h_a)^2+16m_a^2\right]$. Using $r_a=S/(s-a), h_a=2S/a, s=(a+b+c)/2$ and formula (2.31), we easily get

$$(r_b+r_c+2h_a)^2+16m_a^2=\frac{T_a}{a^2(c+a-b)(a+b-c)}, \tag{2.38}$$

where

$$\begin{aligned}T_a=&-8a^6+(20b^2-8bc+20c^2)a^4-(13b^2+6bc+13c^2)(b-c)^2a^2\\&+(b+c)^2(b-c)^4.\end{aligned}$$

Thus we have

$$\prod\left[(r_b+r_c+2h_a)^2+16m_a^2\right]=\frac{\prod T_a}{\prod a^2(b+c-a)^2}. \tag{2.39}$$

By expanding and arranging, one can obtain the following identity:

$$\begin{aligned}\prod T_a =&2118\sum a^{18}+16\sum a\sum a^{17}+132\sum a^2\sum a^{16}\\
&-280\sum a^3\sum a^{15}-549\sum a^4\sum a^{14}+1378\sum a^5\sum a^{13}\\
&+935\sum a^6\sum a^{12}-3248\sum a^7\sum a^{11}-510\sum a^8\sum a^{10}\\
&+4268\sum b^9c^9+abc\Big(-64\sum a^{15}-200\sum a\sum a^{14}\\
&+596\sum a^2\sum a^{13}+394\sum a^3\sum a^{12}-2350\sum a^4\sum a^{11}\\
&+\sum a^5\sum a^{10}\Big)+(abc)^2\Big(2580\sum a\sum a^{11}+4769\sum a^2\sum a^{10}\\
&-1361\sum a^{12}-6620\sum a^3\sum a^9-5350\sum a^4\sum a^8\\
&+4240\sum a^5\sum a^7+4382\sum b^6c^6\Big)+(abc)^3\Big(-9816\sum a^9\\
&+1384\sum a\sum a^8-2768\sum a\sum a^8+11416\sum a^2\sum a^7\\
&-5760\sum a^3\sum a^6-812\sum a^4\sum a^5\Big)+(abc)^4\Big(-17948\sum a^6\\
&+8228\sum a^2\sum a^4-728\sum a\sum a^5-564\sum b^3c^3\Big)\\
&+(abc)^5\Big(904\sum a^3-2000\sum a\sum a^2\Big)+9774(abc)^6.\end{aligned} \tag{2.40}$$

With the help of Maple, using (2.1)-(2.3) and identities given in Lemma 2.1-2.3, we easily get

$$\prod T_a=1024r^6s^4M_0, \tag{2.41}$$

where M_0 is the same as in Lemma 2.4. Also, it is clear that $M_0>0$ from (2.38), (2.39) and (2.41).

It follows from identities (2.39), (2.41) and (2.36) that

$$\prod\left[(r_b+r_c+2h_a)^2+16m_a^2\right]=\frac{64M_0}{R^2}. \tag{2.42}$$

Finally, by identities (2.30), (2.37) and (2.42) we get identity (2.29). This completes the proof of Lemma 2.4.

Lemma 2.5. *For any triangle ABC the following inequality holds:*

$$m_am_bm_c\geq\frac{1}{8R}\sum b^2c^2, \tag{2.43}$$

with equality if and only if triangle ABC is isosceles.

Remark 2.1. *Inequality (2.43) is equivalent to*

$$abcm_am_bm_c\geq\frac{1}{2}S\sum b^2c^2, \tag{2.44}$$

which was first proved by X. Z. Yang in a Chinese paper [13].

Lemma 2.6. *For the acute triangle ABC the following inequality holds:*

$$s^2 \geq 4R^2 - Rr + 13r^2, \tag{2.45}$$

with equality if and only if triangle ABC is equilateral.

Inequality (2.45) was first obtained by X. L. Chen in [12]. Recently, the author [5] obtained the following improvement:

$$s^2 \geq 4R^2 - Rr + 13r^2 + \frac{(R-2r)r^3}{R^2}. \tag{2.46}$$

with equality if and only if $\triangle ABC$ is equilateral or right isosceles.

Lemma 2.7. *For the acute triangle ABC, the following inequality holds:*

$$s^2 \geq 16Rr - 3r^2 - \frac{4r^3}{R}, \tag{2.47}$$

with equality if and only if $\triangle ABC$ is equilateral or right isosceles.

Inequality (2.47) was established by the author in [2] and studied in [5] and [9].

3. Proof of Theorem 1.1

In this sections, we shall prove Theorem 1.1.

Proof. By the simplest arithmetic-geometric mean inequality, we have

$$M_a \equiv \frac{1}{8}(r_b + r_c + 2h_a) + \frac{2m_a^2}{r_b + r_c + 2h_a} \geq m_a,$$

Two similar relations are valid, hence

$$\sum m_b m_c \geq m_a m_b m_c \sum \frac{1}{M_a}.$$

According to Lemma 2.4 and 2.5, so we have

$$\sum m_b m_c \geq \frac{N_0}{RM_0} \sum b^2 c^2. \tag{3.1}$$

Thus, to prove inequality (1.6) we only need to show that

$$\frac{N_0}{RM_0} \sum b^2 c^2 \sum m_a^2 \geq s^4.$$

Note that $M_0 > 0$ and the previous identity (1.2), we see that the above inequality is equivalent to

$$P \equiv \frac{3}{2} N_0 \sum a^2 \sum b^2 c^2 - 2RM_0 s^4 \geq 0. \tag{3.2}$$

With the help of Maple, substituting the expressions of M_0 and N_0 (given in Lemma 2.4) into (3.2), and using (2.4) and (2.17), we easily obtain the

following equivalent inequality:

$$\begin{aligned}P \equiv\ & (69R+294r)s^{12}+(1180R^3+5312R^2r-4088Rr^2\\ & +2208r^3)s^{10}+(-1632R^5-46320R^4r-140088R^3r^2\\ & +29184R^2r^3-18317Rr^4+2454r^5)s^8+(-2048R^7\\ & +8960R^6r+461824R^5r^2+677936R^4r^3-100384R^3r^4\\ & -38048R^2r^5+6688Rr^6-1392r^7)s^6+(4R+r)(8192R^7\\ & +75008R^6r-240640R^5r^2-334896R^4r^3+116680R^3r^4\\ & +49332R^2r^5-16725Rr^6-2766r^7)rs^4-12(1488R^5\\ & -2264R^4r-1603R^3r^2+1376R^2r^3+810Rr^4\\ & +68r^5)(4R+r)^3r^3s^2+3(4R-3r)(28R^2-5Rr\\ & -2r^2)(4R+r)^6r^4\geq 0, \end{aligned} \tag{3.3}$$

which needs to be proved.

Putting $v_0=s^2-4R^2+Rr-13r^2$, then by Lemma 2.6 we have $v_0\geq 0$. After analysis, we obtain the following identity:

$$P=m_6v_0^6+m_5v_0^5+m_4v_0^4+m_3v_0^3+m_2v_0^2+m_1v_0+m_0, \tag{3.4}$$

where

$$\begin{aligned}m_6=&69R+294r,\\ m_5=&2836R^3+11954R^2r-470Rr^2+25140r^3,\\ m_4=&38528R^5+116300R^4r-98313R^3r^2+875204R^2r^3\\ &-234822Rr^4+891264r^5,\\ m_3=&248960R^7+339968R^6r-923768R^5r^2+8555452R^4r^3\\ &-10638708R^3r^4+25404912R^2r^5-8387712Rr^6\\ &+16776096r^7,\\ m_2=&838912R^9-524672R^8r-1756288R^7r^2+32776728R^6r^3\\ &-85585821R^5r^4+217310982R^4r^5-252654676R^3r^6\\ &+365924480R^2r^7-128921408Rr^8+176895072r^9,\\ m_1=&1418240R^{11}-4237824R^{10}r+5652352R^9r^2\\ &+42711040R^8r^3-223887580R^7r^4+696538386R^6r^5\\ &-1416606710R^5r^6+2332451180R^4r^7-2445007456R^3r^8\\ &+2617804832R^2r^9-942182688Rr^{10}+991061568r^{11},\\ m_0=&(R-2r)(942080R^{12}-3650560R^{11}r+8974592R^{10}r^2\\ &+13640576R^9r^3-117774080R^8r^4+454028764R^7r^5\\ &-1161545831R^6r^6+1931362178R^5r^7-3220701970R^4r^8\\ &+2629766332R^3r^9-3339150024R^2r^{10}+768270384Rr^{11}\\ &-1152724608r^{12}).\end{aligned}$$

For any triangle we have the well-known Euler's inequality:

$$R \geq 2r, \tag{3.5}$$

so it is easy to see that $m_5 > 0, m_4 > 0$ and $m_3 > 0$. Putting $R = 2r + e$ $(e \geq 0)$ and substituting it into m_2, expanding gives

$$\begin{aligned} m_2 =& 27328e^7 + 402864e^6r + 2212040e^5r^2 + 6951908e^4r^3 \\ &+ 13875738e^3r^4 + 18180108e^2r^5 + 15670464er^6 + 8377344r^7. \end{aligned} \tag{3.6}$$

So we have $m_2 > 0$.

According to identity (3.4) and inequality $v_0 \geq 0$, for proving inequality inequality (3.3) it remains to show that

$$P_0 \equiv m_1v_0 + m_0 \geq 0. \tag{3.7}$$

A simple calculation shows that the above inequality is equivalent to

$$P_0 \equiv m_1s^2 + j_0 \geq 0, \tag{3.8}$$

where

$$\begin{aligned} j_0 =& -4730880R^{13} + 12834816R^{12}r - 29008640R^{11}r^2 \\ &- 114408704R^{10}r^3 + 719725552R^9r^4 - 2875707720R^8r^5 \\ &+ 7203900407R^7r^6 - 15546956608R^6r^7 + 23444941908R^5r^8 \\ &- 34166921852R^4r^9 + 29572949824R^3r^{10} - 31491321344R^2r^{11} \\ &+ 10550171136Rr^{12} - 10578351168r^{13}. \end{aligned}$$

We set $h_0 = R^2 - 2Rr - r^2$. Next, we shall consider two cases to complete the proof of inequality (3.8).

Case 1. $h_0 \geq 0$.

Firstly, it is easy to obtain

$$\begin{aligned} m_1 =& 1418240e^{11} + 26963456e^{10}r + 232908672e^9r^2 \\ &+ 1253721856e^8r^3 + 4693423908e^7r^4 + 12872300810e^6r^5 \\ &+ 26425692338e^5r^6 + 40846915944e^4r^7 + 47358139536e^3r^8 \\ &+ 41016938112e^2r^9 + 25969102848er^{10} + 10319560704r^{11}, \end{aligned} \tag{3.9}$$

where $e = R - 2r$. So we have the strict inequality $m_1 > 0$.

Note that for any non-obtuse triangle ABC we have Ciamberlini's fundamental inequality (see [1] and [10]):

$$s \geq 2R + r, \tag{3.10}$$

with equality if and only if $\triangle ABC$ is a right-angled triangle. Based on Ciamberlini's inequality, Euler's inequality and the assumption we find the following identity (which can be verified by expanding):

$$P_0 = m_1\left[s^2 - (2R + r)^2\right] + n_1h_0(R - 2r)^2 + n_2h_0 + n_3, \tag{3.11}$$

where

$$\begin{aligned} n_1 =&942080R^9 + 7208960R^8r + 10958592R^7r^2 + 65028352R^6r^3 \\ &+ 302901504R^5r^4 + 232373904R^4r^5 + 2466014187R^3r^6 \\ &+ 2524651260R^2r^7 + 11740963229Rr^8 + 22324886710r^9, \\ n_2 =&(58591817473R - 100005661900r)r^{10}, \\ n_3 =&(8416128149R - 20293404660r)r^{12}. \end{aligned}$$

By Euler's inequality we have $n_2 > 0$. Also, it follows from the assumption $h_0 \geq 0$ that $R \geq (1+\sqrt{2})r$, which implies $n_3 > 0$. Finally, from identity (3.11) we deduce that $P_0 > 0$ strictly holds.

Case 2. $h_0 < 0$.

By Lemma 2.7, we have

$$Rs^2 - 16R^2r + 3Rr^2 + 4r^3 \geq 0. \tag{3.12}$$

Based on this inequality, Euler's inequality and the assumption, we obtain the following identity:

$$\begin{aligned} RP_0 =&m_1(Rs^2 - 16R^2r + 3Rr^2 + 4r^3) + (R-2r)(t_1 - t_2h_0) \\ &- t_3h_0(R-2r)^2 - t_4h_0(R-2r)^3, \end{aligned} \tag{3.13}$$

where

$$\begin{aligned} t_1 =&(8416128149r - 3461148362R)Rr^{11}, \\ t_2 =&7331897854r^{11}, \\ t_3 =&15478915727r^{10}, \\ t_4 =&4730880R^9 + 2320384R^8r + 10821376R^7r^2 + 173135616R^6r^3 \\ &- 225496688R^5r^4 + 1056786056R^4r^5 - 768943451R^3r^6 \\ &+ 2307074550R^2r^7 + 907807679Rr^8 + 6402014184r^9. \end{aligned}$$

It follows from the assumption $h_0 < 0$ that $R < (\sqrt{2}+1)r$, i.e. $r > (\sqrt{2}-1)R$ which implies $8416128149r - 3461148362R > 0$ and then $t_1 > 0$. By Euler's inequality, it is easy to see that $t_4 > 0$. Again, note that $m_1 > 0$ and inequality (3.12). Therefore, from identity (3.13) we immediately conclude that $P_0 \geq 0$, i.e. inequality (3.8) holds under the assumption $h_0 < 0$.

Combining the arguments of the above two cases, we proved inequality (3.8) holds for all acute triangles. Therefore, inequalities (3.3) and (1.6) are proved.

Under Case 2, from identity (3.13) one sees that the equality in $P_0 \geq 0$ holds only when $Rs^2 - 16R^2r + 3Rr^2 + 4r^3 = 0$ and $R - 2r = 0$ which mean $\triangle ABC$ must be equilateral. And we can further deduce that both equalities in (3.3) and (1.6) hold if and only if triangle ABC is equilateral. This completes the proof of Theorem 1.1.

4. Proof of Theorem 1.2

In this section, we shall prove Theorem 1.2.

Proof. We begin with the proved inequality (3.1), which shows that to prove inequality (1.9) we only need to show that

$$\frac{N_0}{RM_0}\sum b^2c^2 \geq \frac{1}{12}\left(\sum h_a + \sum r_a\right)^2. \tag{4.1}$$

Using the following two known identities:

$$\sum r_a = 4R + r, \tag{4.2}$$

$$\sum h_a = \frac{s^2 + 4Rr + r^2}{2R}, \tag{4.3}$$

we easily know that inequality (4.1) is equivalent to

$$Q \equiv\ 48RN_0\sum b^2c^2 - M_0(s^2 + 8R^2 + 6Rr + r^2)^2 \geq 0. \tag{4.4}$$

With the help of Maple, substituting the expressions of M_0 and N_0 (given in Lemma 2.4) into (4.4) and using the previous identity (2.17), we obtain the following equivalent inequality:

$$\begin{aligned} Q \equiv & -9s^{12} + (968R^2 + 4348Rr - 1270r^2)s^{10} + (21104R^4 + 85024R^3r \\ & - 52356R^2r^2 + 31724Rr^3 - 7047r^4)s^8 + (-18176R^6 - 582592R^5r \\ & - 1536768R^4r^2 + 71072R^3r^3 - 103136R^2r^4 + 13272Rr^5 \\ & - 11572r^6)s^6 + (-123904R^8 - 650752R^7r + 2230976R^6r^2 \\ & + 4139456R^5r^3 - 1106704R^4r^4 - 422880R^3r^5 + 126536R^2r^6 \\ & - 27176Rr^7 - 7047r^8)s^4 - 2(2048R^8 - 57344R^7r - 338560R^6r^2 \\ & - 140480R^5r^3 - 27408R^4r^4 - 122928R^3r^5 - 46428R^2r^6 \\ & + 1714Rr^7 + 635r^8)(4R + r)^2s^2 + (4R - 3r)(256R^5 - 1504R^3r^2 \\ & + 220R^2r^3 + 116Rr^4 + 3r^5)(4R + r)^5r \geq 0, \end{aligned} \tag{4.5}$$

which needs to be proved.

We now set

$$\begin{aligned} v_0 &= s^2 - 4R^2 + Rr - 13r^2, \\ t_0 &= -s^4 + (4R^2 + 20Rr - 2r^2)s^2 - r(4R + r)^3. \end{aligned}$$

By Lemma 2.3 we have $v_0 \geq 0$. For any triangle we have the following fundamental triangle inequality (cf. [10] and [8]):

$$s^4 - (4R^2 + 20Rr - 2r^2)s^2 + r(4R + r)^3 \leq 0, \tag{4.6}$$

hence $t_0 \geq 0$. Based on $v_0 \geq 0$ (from Lemma 2.6), $t_0 \geq 0$ and Euler's inequality $R \geq 2r$, after analysis we obtain the following identity:

$$Q = 9t_0^3 + (k_3v_0^3 + k_2v_0^2 + k_1v_0 + k_0)v_0^2 + p_0s^2 + q_0, \tag{4.7}$$

where

$$\begin{aligned}
k_3 =&860R^2+3808Rr-1216r^2,\\
k_2 =&38736R^4+162932R^3r-28152R^2r^2+283488Rr^3-85952r^4,\\
k_1 =&463424R^6+1306432R^5r-1150280R^4r^2+9208992R^3r^3\\
&-3249568R^2r^4+8344832Rr^5-2425856r^6,\\
k_0 =&2268928R^8+1905984R^7r-8027168R^6r^2+67031304R^5r^3\\
&-88216976R^4r^4+194510368R^3r^5-79382336R^2r^6\\
&+121660160Rr^7-34175488r^8,\\
p_0 =&4657152R^{10}-8574464R^9r-5380928R^8r^2+152026432R^7r^3\\
&-477513236R^6r^4+1087182016R^5r^5-1366891936R^4r^6\\
&+1820898432R^3r^7-786611520R^2r^8+879826432Rr^9\\
&-240350208r^{10},\\
q_0 =&-15679488R^{12}+21075968R^{11}r-10592768R^{10}r^2\\
&-438998656R^9r^3+1587890976R^8r^4-5113408720R^7r^5\\
&+8944375308R^6r^6-17107582688R^5r^7+16242029088R^4r^8\\
&-21599581568R^3r^9+9240900288R^2r^{10}-9150137344Rr^{11}\\
&+2449434624r^{12}.
\end{aligned}$$

Note that Euler's inequality $R\geq 2r$. It is easily seen that $k_3>0, k_2>0$ and $k_1>0$. Letting $R=2r+e(e\geq 0)$ and substituting it into k_0, we get

$$\begin{aligned}
k_0 =&2268928e^8+38208832e^7r+272776544e^6r^2+1147287688e^5r^3\\
&+3175340864e^4r^4+6018949856e^3r^5+7753196672e^2r^6\\
&+6313863168er^7+2492301312r^8>0.
\end{aligned}\tag{4.8}$$

Therefore, to prove inequality $Q\geq 0$ it remains to prove that

$$Q_0\equiv p_0s^2+q_0\geq 0.\tag{4.9}$$

We still set $h_0=R^2-2Rr-r^2$ and consider the following two cases to complete the proof of inequality (4.9).

Case 1. $h_0\geq 0$.

We first show that $p_0>0$. Putting $R=2r+e$, then it is easy to get

$$\begin{aligned}
p_0 =&4657152e^{10}+84568576e^9r+678566080e^8r^2+3302074688e^7r^3\\
&+10934183788e^6r^4+25985742032e^5r^5+45414779696e^4r^6\\
&+58900641024e^3r^7+56363157504e^2r^8+37822464000er^9\\
&+13759414272r^{10},
\end{aligned}\tag{4.10}$$

which shows $p_0>0$ since $e\geq 0$.

Under the assumption $h_0\geq 0$, we have $R\geq(\sqrt{2}+1)r$. Letting $H_0=R-(\sqrt{2}+1)r$, then $H_0\geq 0$. After analysis we obtain the following identity:

$$Q_0=p_0\left[s^2-(2R+r)^2\right]+f_1(R-2r)h_0+8f_2H_0+f_3,\tag{4.11}$$

where

$$\begin{aligned}
f_1 =& 2949120R^9 + 17203200R^8r - 1791744R^7r^2 + 74334080R^6r^3 \\
&+ 548867984R^5r^4 - 546653984R^4r^5 + 3366134344R^3r^6 \\
&- 197611328R^2r^7 + 9216744424Rr^8 + 11321719504r^9, \\
f_2 =& [3630507633R - (4073287255 - 3630507633\sqrt{2})r]r^{10}, \\
f_3 =& 8(633433687 - 442779622\sqrt{2})r^{12}.
\end{aligned}$$

By Euler's inequality we know that $f_1 > 0$ and $f_2 > 0$. Note that $p_0 > 0$ and $f_3 > 0$. From identity (4.11), by Ciamberlini's inequality (3.10) and Euler's inequality we deduce that $Q_0 > 0$ holds. Therefore, we proved that inequality (4.9) is a strict inequality under Case 1.

Case 2. $h_0 < 0$.

In this case, we have $R \leq (\sqrt{2}+1)r$ and $H_0 \equiv R - (\sqrt{2}+1)r \leq 0$. After analysis we obtain the following identity:

$$\begin{aligned}
RQ_0 =& p_0(Rs^2 - 16R^2r + 3Rr^2 + 4r^3) \\
&+ 8(R-2r)\left[-f_4h_0 + (-61471492H_0 + f_5)r^{11}\right],
\end{aligned} \tag{4.12}$$

where

$$\begin{aligned}
f_4 =& 1959936R^{10} - 4109056R^9r - 2096576R^8r^2 + 64770832R^7r^3 \\
&- 235253148R^6r^4 + 517390074R^5r^5 - 749687117R^4r^6 \\
&+ 960755952R^3r^7 - 583738957R^2r^8 + 554712230Rr^9 \\
&- 250741617r^{10}, \\
f_5 =& (129182573 - 61471492\sqrt{2})r > 0.
\end{aligned}$$

Letting $R = 2r + e$, then we easily get

$$\begin{aligned}
f_4 =& 1959936e^{10} + 35089664e^9r + 276728896e^8r^2 + 1321060112e^7r^3 \\
&+ 4260821316e^6r^4 + 9716903146e^5r^5 + 15870518631e^4r^6 \\
&+ 18547291928e^3r^7 + 15531623931e^2r^8 + 9419487794er^9 \\
&+ 3372142487r^{10},
\end{aligned}$$

hence $f_4 > 0$. So we have $-f_4h_0 + (-61471492H_0 + f_5)r^{11} > 0$ under Case 2. By Lemma 2.7 we have $Rs^2 - 16R^2r + 3Rr^2 + 4r^3 \geq 0$. Therefore, from identity (4.12) we conclude that $Q_0 \geq 0$ i.e. inequality (4.9) holds for the non-obtuse triangle ABC under Case 2. Furthermore, we immediately determine that the equality in $Q_0 \geq 0$ occurs if and only if triangle ABC is equilateral under Case 2.

Combining the arguments of the above two cases, we proved that inequality (4.9) holds for all acute triangles. So, inequalities (4.5), (4.1) and (1.9) are proved. Moreover, it is to determine that equality in (1.9) holds if and only if triangle ABC is equilateral. This completes the proof of Theorem 1.2.

5. Open problems

In the last section, we propose three conjectures as open problems.

Inspired by inequality (1.6), the author found and proved that the following inequality

$$\sum m_a \sum m_a^3 \geq s^4 \tag{5.1}$$

holds for any triangle.

We denote by w_a, w_b, w_c the three angle-bisectors of triangle ABC. Note that $m_a \geq w_a$ etc., we conjecture that the above inequality can be improved to the following:

Conjecture 5.1. *In any triangle ABC the following inequality holds:*

$$\sum w_a \sum m_a^3 \geq s^4. \tag{5.2}$$

For the acute triangle ABC, the author also proved the following inequality:

$$\sum h_a \sum m_a^3 \geq s^4. \tag{5.3}$$

A similar conjecture is

Conjecture 5.2. *In the acute triangle ABC the following inequality holds:*

$$\sum m_a \sum w_a^3 \geq s^4. \tag{5.4}$$

From Theorem 1.2, we easily deduce that the following linear inequality

$$2\sum m_a \geq \sum h_a + \sum r_a \tag{5.5}$$

holds for the acute triangle ABC. We propose here the following stronger inequality:

Conjecture 5.3. *In the acute triangle ABC the following inequality holds:*

$$2\left(\sum m_a\right)^5 \geq \left(\sum h_a\right)^5 + \left(\sum r_a\right)^5. \tag{5.6}$$

If this inequality holds, then inequality (5.5) follows immediately by using the power mean inequality.

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